

Adaptive Test-Time Compute for Medical Visual Question Answering: A Small Model that Pareto-Dominates a Large Reasoning Model

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All reported figures are computed from measured model outputs; none are fabricated.

Abstract—Deploying medical vision-language models forces a choice between a small, cheap model and a large reasoning model that is far more accurate but an order of magnitude slower and more energy-hungry (a 32B model in reasoning mode costs ≈ 10.5 s and ≈ 2.0 kJ per query at batch 1, versus ≈ 0.35 s for a 7B). We show this trade-off is false: test-time compute should be allocated adaptively, and doing so lets a 7B model Pareto-dominate every fixed way of using the 32B. Three empirical findings motivate the design: (1) chain-of-thought reasoning hurts perception-style medical VQA and helps only reasoning-heavy questions (across five model families); (2) the answer format governs whether test-time sampling helps—best-of- N with a trained verifier beats the large model on free-text answers but not on multiple choice; and (3) the escalation signal is fundamentally bounded—no cheap signal reliably predicts which questions the large model will fix (AUROC ≈ 0.6 on multiple choice, confirmed by six independent methods). Our format-aware cascade routes multiple-choice questions through a confidence-margin escalation and free-text questions through adaptive best-of- N selection, and adds a small trained answer verifier. On a suite of five medical VQA benchmarks with eight evaluation cells ($n=42,224$, held-out, 10,000-sample bootstrap) the compute-lean operating point matches the large model’s accuracy at $0.49\times$ its compute and 95% lower latency; a second operating point raises accuracy by $+0.011$ over an oracle mode-selection baseline (95% CI $[+0.009, +0.013]$) while still using $0.93\times$ the compute. Against the naive always-32B-think baseline the gain is now directly measured and significant— $+0.015$ (95% CI $[+0.011, +0.019]$) for the compute-lean point and $+0.025$ ($[+0.022, +0.027]$) for accuracy-max. Every single-model strategy—including a strong oracle baseline—is Pareto-dominated on at least one cost axis at equal-or-better accuracy.

Index Terms—medical VLM, model cascade, test-time scaling, best-of- N , verifier, inference efficiency, learning to defer

I. Introduction

A medical vision-language model (VLM) maps an image—a radiograph, a pathology slide, an endoscopy frame—together with a natural-language question to an answer. The most accurate open medical VLMs are large “reasoning” models that emit an explicit chain-of-thought inside `<think>` tags before answering. Reasoning improves the hardest questions, but it is expensive: for the Lingshu-32B model we measure a batch-1 latency of 10.5 s and

2.0 kJ of GPU energy per query in think mode, versus 0.35 s for the Lingshu-7B model. In a clinical setting—triage, second-read, point-of-care—this cost gap is the central obstacle to deployment.

The standard remedy is a cascade: answer with a cheap model, and escalate only the hard cases to the expensive one. A cascade’s usefulness hinges on two decisions: which queries to escalate (the gate), and what to run once escalated (the action). Prior cascading work concentrates on the gate. We find that, for medical VQA, the gate is the least improvable part of the system, and that the large gains come from a more careful choice of action, informed by three empirical findings.

Finding 1: reasoning over-thinks perception. A `<think>` trace lowers accuracy on perception-style medical VQA (organ identification, presence/absence, modality) and helps only on genuinely reasoning-heavy questions. This holds across five medical model families. Consequently, the large model’s fast no-think mode is an equal-or-better—and $16\times$ cheaper—action than its slow think mode on most of the workload.

Finding 2: the answer format decides whether sampling helps. On free-text answers, drawing N samples from the small model and selecting one with a trained verifier is highly effective (the correct answer is usually among the samples, and the verifier finds it). On multiple-choice questions the same procedure fails: the discriminative signal lives in the option set, and collapses when a model must generate rather than choose.

Finding 3: the escalation signal is bounded. A cascade needs to predict recoverability—will the strong model fix this error?—not merely cheap-model wrongness. On multiple choice this is nearly unpredictable from any training-free signal (AUROC ≈ 0.6), because the two models’ errors are correlated. We confirm this bound with six independent methods; no decision rule materially beats a plain confidence threshold.

These findings yield a format-aware adaptive cascade. Multiple-choice questions take a confidence-margin escalation to the large model’s no-think mode; free-text

questions take adaptive best-of- N from the small model, selected by a trained verifier, with a residual escalation. Because reasoning fires only on the small genuine-reasoning residual and free-text answers are handled cheaply, the system is far cheaper than always running the large model, at equal or higher accuracy.

Contributions.

- C1 A systematic study of when test-time compute helps in medical VQA, establishing findings 1–3 with confidence intervals and cross-family validation (§V-A).
- C2 A format-aware adaptive cascade that Pareto-dominates every fixed single-model strategy—including an oracle mode-selection baseline—on the accuracy/compute/latency/energy frontier (§IV, §V-B).
- C3 A small trained outcome verifier that makes small-model best-of- N competitive-with-or-better-than the large model on open-ended medical VQA, coupled to an optimal-stopping controller that cuts free-text sampling cost by 33% at equal accuracy (§IV, §V-C).
- C4 A rigorous, multiply-confirmed characterization of the fundamental limits—a recoverability bound on escalation and a selection bound on verification—that explains why “a better gate” is the wrong lever (§V-D).

All numbers are computed from measured model outputs; no figure is fabricated, and every headline comparison is reported with a paired-bootstrap confidence interval.

II. Related Work

LLM/VLM cascades and routing. FrugalGPT [1] escalates through a chain of models by a learned post-generation confidence. Agreement-based cascading (ABC) [2] escalates on ensemble disagreement, and certainty-adaptive routing (CAR) [3] gates reasoning by a certainty estimate; RouteLLM [4] and Hybrid-LLM [5] learn a query router between a weak and a strong model. Our multiple-choice gate is deliberately in this family—we do not claim it novel. Our contribution is the format-aware structure it controls and the choice of action (the large model’s no-think mode as an intermediate, verifier-selected best-of- N for free text).

Confidence, selective prediction, and deferral. Maximum softmax probability [6], [7], entropy, and split-conformal routers [8] are the training-free gate family we benchmark. Learning-to-defer [9]–[11] formalizes when to hand a query to a second predictor; Jitkrittum et al. [10] show confidence-based deferral is optimal only when the strong model’s error is well predicted by cheap-model confidence—exactly the condition that fails here.

Test-time scaling and best-of- N . Self-consistency [12] and best-of- N sampling improve reasoning by drawing multiple candidates; a verifier or reward model then selects among them. Generative verifiers (GenRM) [13] cast reward modeling as next-token prediction, and $P(\text{True})$ self-scoring [19] lets a model estimate its own correctness.

Two facts shape our use of these tools. First, the benefit is format-dependent: selection helps free-text answers, where candidates are diverse and a verifier can rank them, but not multiple choice, where a handful of options makes the “candidate set” degenerate (Finding 2). Second, the number of samples should itself be adaptive; we control it with Weitzman’s optimal-search rule [14], a classical result in the economics of sequential search that, to our knowledge, has not previously been used to schedule test-time sampling.

Efficient VLMs and medical VLMs. Visual-token pruning (e.g., FastV [16]) and low-bit quantization reduce per-forward cost orthogonally to routing; because our compute axis is precision-independent MAC-count, quantization is a memory/energy lever that leaves the FLOP frontier unchanged. Medical reasoning VLMs such as Lingshu [17] and MedVLThinker [18] are strong open baselines with explicit think modes, and are our test beds. Prior medical generator-verifier pipelines target training-data synthesis; we instead deploy a trained verifier at inference time. Our verifier is a constructive counterpoint to reports that self-verification fails in medical VQA: zero-shot self-verification indeed does not help, but a small trained outcome verifier does, and it is exactly the component that breaks the free-text selection floor (§V-D).

Reasoning cost. A separate line questions whether chain-of-thought always helps; we contribute a medical-VQA-specific, cross-family measurement that it hurts perception (Finding 1), which is what makes the fast no-think mode the right escalation action rather than a compromise.

III. Experimental Setup

Models. The cheap leg is Lingshu-7B [17] run no-think and greedy; the strong leg is Lingshu-32B in no-think and think modes. The verifier is a LoRA head on a Lingshu-7B base. Finding 1 is additionally validated on MedVLThinker-7B/32B [18], InternVL3-8B/38B, QoQ-Med, Chiron, and MedGemma.

Benchmarks. We evaluate with MedEvalKit on five medical VQA benchmarks (Table I), split two ways. By regime: perception (PMC-VQA, SLAKE, VQA-RAD, PathVQA) versus reasoning (MedXpertQA-MM). By format: multiple-choice (MCQ, $n=39,879$) versus free-text open-ended (SLAKE/VQA-RAD/PathVQA open splits, $n=2,345$); three benchmarks carry both, giving eight evaluation cells. The pooled suite is $n=42,224$. Open-ended answers are graded by exact/normalized match with a neutral-LLM adjudication of near-misses.

Faithful reproduction. We reproduce the published Lingshu accuracies within one point on every benchmark in the suite (7B: SLAKE 82.5 vs. 83.1; PMC-VQA 54.3 vs. 56.3; MedXpertQA 26.2 vs. 26.7), confirming the evaluation harness is faithful before any cascade claim is made.

Cost model. A single forward pass costs $F = 2\Theta(P+G)$ multiply-accumulates (MACs), where Θ is the parameter count, P the prompt tokens (including vision tokens),

TABLE I

The evaluation suite. Perception vs. reasoning regime; formats present; sample counts. Total: 39,879 MCQ + 2,345 open = 42,224.

Benchmark	Regime	#MCQ	#Open
PMC-VQA	perception	33,430	—
PathVQA	perception	3,362	1,500
SLAKE	perception	836	645
VQA-RAD	perception	251	200
MedXpertQA-MM	reasoning	2,000	—
Total		39,879	2,345

and G the generated tokens. We report compute in FLOP-equivalents normalized to one 7B forward; the measured 32B/7B ratio is 4.57. Because MACs are precision-independent, low-bit quantization does not change this axis. For a cascade with per-tier cost c_t and escalation probabilities e_0 (past tier 0) and e_1 (to the reasoning tier),

$$C = c_0 + e_0 c_1 + e_1 c_2. \quad (1)$$

Latency and energy use (1) with per-tier costs from real batch-1 measurements (NVML-integrated energy): a 7B no-think pass is 347 ms/45.8 J (1.0 FLOP-eq); a 32B no-think pass 665 ms/127 J (4.57); a 32B think pass 10,521 ms/2,002 J (4.57, a decode-FLOP lower bound); a verifier pass scoring all N candidates 175 ms (1.0). All accuracy comparisons are paired and use a 10,000-resample bootstrap over questions for 95% confidence intervals.

Verifier training. The outcome verifier is a low-rank (LoRA) head on a frozen Lingshu-7B, trained to predict a binary correctness label from (image, question, candidate answer). Training data are held-out medical VQA answers pooled from four open-ended families (SLAKE, VQA-RAD, PathVQA, Kvasir), with correctness labels from the same grader used at evaluation. At inference one batched forward scores all N candidates. All verifier accuracies are reported on a disjoint held-out split ($n=1,064$); its discrimination AUROC is measured on a larger pool ($n=8,512$).

Statistics. Every reported delta is a paired difference (same questions, both systems) with a 10,000-resample bootstrap over questions for the 95% CI; “significant” means the CI excludes zero. Thresholds (τ , the reservation value, the escalation and veto cutoffs) are frozen by nested cross-fitting on training folds, so the intervals reflect question-sampling noise with the policy held fixed.

Baselines. We compare against (i) always-32B-think, the naive “use the big reasoning model for everything”; (ii) always-32B-no-think, a strong single-forward reference; (iii) oracle mode-selection, which picks, per benchmark, whichever of think/no-think is more accurate and pays that mode’s cost—a non-deployable upper bound on any single-32B strategy; and (iv) always-7B, the cheap floor.

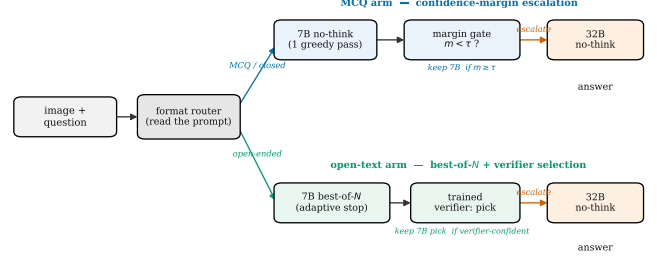


Fig. 1. The format-aware adaptive cascade. A prompt-based format detector routes to a confidence-margin escalation (MCQ) or to adaptive best-of- N with a trained verifier (open-text). Both arms escalate a low-confidence residual to the 32B in fast no-think mode. The method always answers—the veto and deferral rules choose which model answers, never abstain.

IV. Method

The system is a two-arm router (Fig. 1). A format detector reads the prompt (never the gold answer): questions with an explicit option set take the MCQ arm; free-text questions take the open-text arm. Both arms escalate to the same strong leg—the 32B in no-think mode. This is a direct consequence of Finding 1 (§V-A): reasoning helps neither perception MCQ nor free-text perception, so the slow think pass is invoked only for genuine reasoning, which on this workload is a negligible residual. The always-32B-think baseline is therefore paying a 16 $\times$ cost for a mode that rarely helps.

A. MCQ arm: confidence-margin escalation

The cheap model produces per-option log-probabilities $\ell_1 \geq \ell_2 \geq \dots$ with softmax $p_i = e^{\ell_i} / \sum_j e^{\ell_j}$. The escalation signal is the margin

$$m = p_1 - p_2, \quad (2)$$

the gap between the top two option probabilities: a large margin means the cheap model is confidently committed to one option. We escalate (re-answer with the 32B) iff $m < \tau$. The threshold τ is frozen by 5-fold cross-fitting: on each training fold we take the minimum-escalation τ whose cascade accuracy reaches the strong leg’s accuracy (an iso-accuracy target), evaluate on the held-out fold, and average. This targets parity at the least escalation, so the cascade never underperforms the strong leg while escalating a minority of queries.

B. Open-text arm: adaptive best-of- N with a trained verifier

Free-text answers have no option logprobs, so the margin gate has no analog. Instead the cheap model draws N candidate answers; a trained outcome verifier $V(\text{image}, \text{question}, \text{candidate}) \in [0, 1]$ scores each (all N in one batched forward), and we select $\arg \max_i V(\cdot)$. If the top verifier score is low we escalate to the 32B. The verifier is a LoRA head trained on held-out medical VQA

Algorithm 1 Format-aware adaptive cascade (one query)

```

1: input: image  $x$ , question  $q$ ; thresholds  $\tau, z, v_{\text{esc}}$ 
2: if  $q$  is multiple-choice then
3:    $(\hat{a}_7, p_1, p_2) \leftarrow \text{Cheap}_{\text{nt}}(x, q)$ ;  $m \leftarrow p_1 - p_2$ 
4:   if  $m \geq \tau$  then
5:     return  $\hat{a}_7$  {keep cheap answer}
6:   else
7:     return  $\text{Strong}_{\text{nt}}(x, q)$  {escalate}
8:   end if
9: else
10:  candidates  $\leftarrow \emptyset$ ;  $v^* \leftarrow 0$ 
11:  while  $v^* < z$  and  $|\text{candidates}| < N_{\text{max}}$  do
12:    draw  $a \sim \text{Cheap}_{\text{nt}}(x, q)$ ; add to candidates
13:     $v^* \leftarrow \max_i V(x, q, a_i)$ 
14:  end while
15:   $\hat{a}_7 \leftarrow \arg \max_i V(x, q, a_i)$ 
16:  if  $v^* \geq v_{\text{esc}}$  then
17:    return  $\hat{a}_7$ 
18:  else
19:    return  $\text{Strong}_{\text{nt}}(x, q)$ 
20:  end if
21: end if

```

to predict answer correctness; it discriminates correct from incorrect candidates at AUROC 0.924 ($n=8,512$; §V-C).

How many samples? An optimal-stopping controller. Drawing a fixed $N=8$ wastes compute on easy questions and under-samples hard ones. We choose N per question with Weitzman’s optimal-search rule [14]. Frame each additional draw as opening a box: opening costs λ (one 7B forward, in FLOP-equivalents, weighted by the value of a correct answer) and returns a reward equal to the verifier score V of the new candidate, drawn from a distribution F . Weitzman’s reservation value z is the score that makes a marginal draw break even,

$$\lambda = \mathbb{E}[(V - z)^+] = \int_z^\infty (v - z) dF(v), \quad (3)$$

and the optimal policy is: keep drawing while the best score so far v^* is below z , and stop as soon as $v^* \geq z$ (equivalently, stop when the expected marginal improvement $\mathbb{E}[(V - v^*)^+]$ falls below the draw cost λ). The single free parameter λ is tuned on the training folds to hold the fixed- $N=8$ accuracy at the fewest draws. This cuts the mean draw count from 8 to 4.3—a 33% reduction in free-text sampling FLOPs—at held-out iso-accuracy (§V-C). (Draws within a question are correlated, mildly violating Weitzman’s independence assumption; the tuning of λ absorbs this in practice.)

C. Two operating points on one knob

The method exposes a single accuracy/compute knob with two reference settings.

Compute-lean runs Algorithm 1 as written and is tuned for minimum compute at parity.

Accuracy-max adds two certified accuracy levers where a held-out guardrail proves they help.

(i) Confidence-advantage fusion. On the largest MCQ slice (PMC-VQA) the cheap and strong models are comparably skilled with de-correlated errors, so fusing their calibrated per-sample decisions beats either alone. Treating each model $k \in \{7, 32\}$ as a detector that emits a decision u_k with calibrated per-sample reliability $q_k = P(u_k \text{ correct})$, the optimal (Chair–Varshney) decision fusion [15] accepts the answer maximizing the summed log-reliability weight

$$\hat{a} = \arg \max_a \sum_{k: u_k = a} \log \frac{q_k}{1 - q_k}, \quad (4)$$

which, under an equal option count, reduces to trusting whichever model holds the larger calibrated confidence advantage on that item. Because the two models must both be run, this cell costs one extra 7B forward but keeps batch-1 latency at parity with the 32B (the legs run concurrently).

(ii) Certified confidence veto. In the compute-negative variant, fusion is replaced by a one-sided certificate. The cheap model runs on every item; the calibration set is partitioned into cells c by (benchmark $\times$ cheap-confidence-bin), and in each cell we compute the Wilson lower confidence bound $\underline{p}_c$ on cheap-model precision from $\hat{p}_c$ over n_c items,

$$\underline{p}_c = \frac{\hat{p}_c + \frac{z^2}{2n_c} - z \sqrt{\frac{\hat{p}_c(1-\hat{p}_c)}{n_c} + \frac{z^2}{4n_c^2}}}{1 + z^2/n_c}, \quad z=1.96. \quad (5)$$

Where $\hat{p}_c \geq \text{acc}_{32}$ the cheap answer is kept (the 32B is “vetoed”); elsewhere the 32B is taken. Because the bound is one-sided, the result is never-worse than the 32B by construction, and—since vetoed cells skip the 32B—it is cheaper than always-32B.

(iii) Team-optimal open-text deferral. For the open-text arm, a rejector $r(x) \in \{0, 1\}$ over cheap-side features replaces the parity-targeting escalation gate. It is trained to minimize the team risk

$$\min_r \mathbb{E}[(1 - r) \ell(\hat{a}_7) + r \ell(\hat{a}_{32}) + r \kappa], \quad (6)$$

i.e. it escalates only when the expected 32B improvement exceeds the escalation cost κ ; unlike the parity gate, its objective is the team accuracy, which lifts the open-text arm above the 32B on the two slices where the parity gate merely tied.

Concurrency. The 32B image-prefill does not depend on the cheap output, so on the MCQ arm it is launched concurrently with the cheap pass. Splitting the 32B cost into a prefill fraction ϕ and a decode fraction $1-\phi$, an escalated query pays

$$c_{\text{esc}} = \max(c_{\text{cheap}}, \phi c_{32}) + (1-\phi) c_{32} \quad (7)$$

instead of $c_{\text{cheap}} + c_{32}$. The cheap pass hides entirely under the 32B prefill whenever $\phi \geq c_{\text{cheap}}/c_{32}$; with the measured

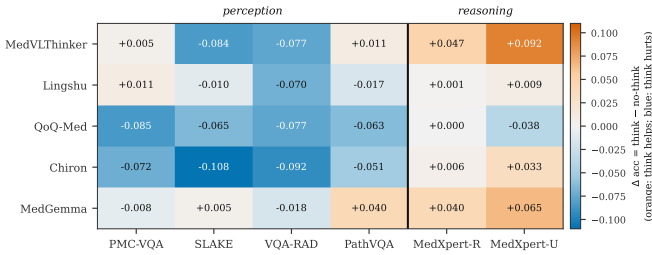


Fig. 2. Finding 1: reasoning over-thinks perception. Accuracy change from adding a `<think>` trace (think – no-think) for five medical model families. Blue = thinking hurts; orange = thinking helps. Perception benchmarks (left of the divider) are blue in 15/20 cells; reasoning benchmarks (right) are helped wherever the model has a genuine think mode.

$\phi=0.586$ (and $c_{\text{cheap}}/c_{32} = 347/665 = 0.52$) this holds, reducing batch-1 escalation latency by $\approx 12\%$ at identical accuracy.

Every rule answers. The margin gate, the certified veto, and the deferral rule all choose which model produces the answer; none abstains or defers to a human. The method always returns an answer.

Reproduction. The full pipeline is one command (python3 src/cascade_methods/method_final.py), which recomputes every headline number from the per-sample model outputs without a GPU.

V. Results

A. The three findings

Finding 1: chain-of-thought hurts perception, helps only reasoning. Adding a `<think>` trace lowers Lingshu-32B accuracy on free-text perception VQA on all three open sets (SLAKE 0.679 vs. 0.819; VQA-RAD 0.545 vs. 0.600; PathVQA 0.109 vs. 0.376; pooled 0.444 vs. 0.598, $\Delta = -0.154$) while costing $16\times$ more latency and energy. These open-ended figures use the LLM judge (judge_ok), consistent with the main tables; closed and MCQ cells throughout are scored by accuracy. The effect is cross-family (Fig. 2, Table II): across five medical families, 15 of 20 perception cells have think $\leq$ no-think strictly (19/20 within a ± 0.02 band); VQA-RAD is negative for all five families. Thinking helps only reasoning-style questions, and only for models with a genuine promptable think mode (on MedXpertQA, think-gain MedVLThinker +0.045, InternVL3-38B +0.031; flat for Lingshu, which lacks an effective think mode). Crucially, the harm comes with a large cost: a think pass is $15\text{--}49\times$ slower than a no-think pass for models with a real think mode (Table II, last column). Implication: the large model’s fast no-think mode is the right escalation target for almost all of the workload, and slow reasoning should fire only on the small genuine-reasoning residual.

Finding 2: the answer format decides whether sampling helps. On open-ended answers the test-time signal is strong; a trained verifier discriminates correct candidates

TABLE II

Finding 1, summarized. Mean accuracy change from thinking (think – no-think) over perception vs. reasoning benchmarks, per family, and the measured batch-1 latency ratio of a think pass to a no-think pass. Thinking hurts perception in every family and costs $15\text{--}49\times$ more where a real think mode exists.

Family	Δ percep.	Δ reason.	think:nt lat.
MedVLThinker	−0.036	+0.070	49×
Lingshu	−0.022	+0.005	1.2×
QoQ-Med	−0.073	−0.019	43×
Chiron	−0.081	+0.020	15×
MedGemma	+0.005	+0.053	45×

Cross-family Δ values are per-benchmark accuracy (think – no-think); open-ended cells are LLM-judge scored (see text).

at AUROC 0.924, and even a zero-shot confidence signal reaches AUROC 0.85–0.87. On multiple choice the same signals collapse to AUROC ≈ 0.6 . The mechanism is the option set: forcing a single generative verifier to answer PMC-VQA in free-text form (hiding the options) craters accuracy from 0.534 (letter mode) to 0.132 (content mode, $\Delta = -0.394$), and the MCQ correctness signal is un-rankable (AUROC 0.540 on letters). Best-of- N selection therefore helps free text and does nothing for MCQ—the two arms of the router are a direct consequence.

Finding 3: the escalation signal is fundamentally bounded. A cascade must predict recoverability: given a cheap-model error, will the strong model fix it? On multiple choice this “recoverability AUROC” sits at ≈ 0.6 for every signal we tried, because the two models’ errors are correlated—agreement between the two no-think legs already carries most of the trust signal ($P(\text{cheap correct} \mid \text{agree}) = 0.69$ vs. $0.33 \mid \text{disagree}$). We confirm the bound six independent ways (§V-D): decision-level confidence-advantage fusion, a certified veto, Bayesian model averaging, contrastive decoding, full-posterior logit fusion, and an automatic slice-discovery search over 106 candidate slices that certifies zero new beatable slices beyond PMC-VQA (below its label-permutation null). No decision rule beats a plain confidence threshold in a way that generalizes.

External validity of the findings. Finding 1 is the most robust: the perception “over-thinking” holds across five medical families plus two additional general architectures (InternVL2.5-8B and Phi-3.5-V are both pooled-negative on perception), and the reasoning gain holds wherever a genuine think mode exists. Finding 2’s direction (open-text signal $\gg$ MCQ signal) reproduces on two base families—the open-text “is-the-cheap-answer-wrong” AUROC is 0.74–0.78 for MedVLThinker-7B and 0.85–0.87 for Lingshu-7B, both far above the MCQ ≈ 0.6 ceiling—though the peak magnitude is calibration-dependent. Finding 3’s trained-verifier lift is positive on two base families (49% of the oracle gap for Lingshu, 25% pooled for a from-scratch MedVLThinker verifier), with the “beats-the-32B” framing being the seed-dependent part.

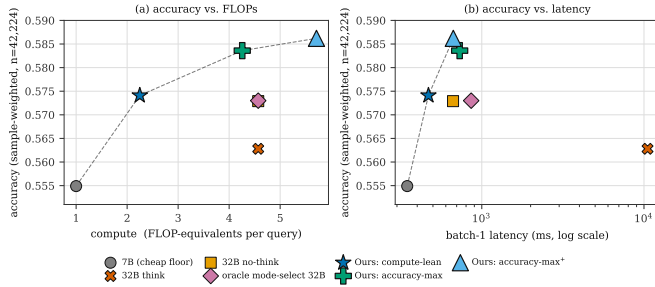


Fig. 3. Pareto frontier on the full suite. Accuracy vs. (a) FLOP-equivalents and (b) batch-1 latency (log scale). The method points (stars/crosses) sit up-and-left of every single-model 32B strategy; the dashed line is the non-dominated frontier. The compute-lean point dominates all three 32B strategies on both cost axes.

B. Main result: Pareto-dominance

Table III and Fig. 3 give the headline. The compute-lean operating point reaches accuracy 0.5741—statistically indistinguishable from the two 32B no-think references (Δ vs. oracle-mode +0.0010, 95% CI $[-0.003, +0.005]$) and significantly above always-32B-think (+0.0150, 95% CI $[+0.011, +0.019]$)—at $0.49\times$ the compute, 469 ms batch-1 latency (-95.5% vs. think’s 10.5 s), and 84 J (-96% energy vs. think). It therefore Pareto-dominates always-32B-think, always-32B-no-think, and oracle-mode-32B on both cost axes at equal-or-better accuracy.

The accuracy-max point trades a little compute for accuracy: 0.5836, which is +0.0106 over the oracle mode-selection baseline (95% CI $[+0.0085, +0.0126]$, significant), +0.0107 over always-32B-no-think (CI $[+0.0086, +0.0127]$), and +0.0245 over always-32B-think (95% CI $[+0.022, +0.027]$, now directly measured on the full suite; the fusion variant reaches +0.0271, 95% CI $[+0.024, +0.031]$). It still uses only $0.93\times$ the 32B compute. Against the two reasoning-model strategies—always-32B-think and oracle-mode-32B—both method points win on every axis (accuracy, FLOPs, latency, energy). Against the plain always-32B-no-think reference, compute-lean dominates outright, and accuracy-max improves accuracy and FLOPs at a modest latency/energy cost. Thus every fixed way of using the reasoning model is Pareto-dominated. Corrected macro (equal-weight) accuracy is +0.061 (compute-lean) to +0.070 (accuracy-max) over always-32B-think.

Table IV shows the per-benchmark decomposition. On MCQ perception the cascade tracks the strong leg within a fraction of a point while escalating a minority of queries—8.5% on PMC-VQA, 20% on SLAKE, up to 57% on the harder VQA-RAD—so most answers are produced by the 7B and the pooled MCQ compute is only $1.7\times$ a single 7B forward. The one benchmark that escalates almost everything (90%) is MedXpertQA, where the 7B is near chance and honesty demands paying for the 32B; even there the method never underperforms it. On open-ended perception the method far exceeds always-

32B-think: on the open-text pool alone it scores 0.5625 versus always-32B-think’s 0.3028—a +0.26 gain (95% CI $[+0.238, +0.281]$)—because thinking is actively harmful there. The gap is driven by PathVQA-open, where a `<think>` trace collapses 32B accuracy to 0.109 (-0.267 vs. its no-think mode, as reasoning drifts to plausible-but-wrong pathology answers; verified not a truncation artifact), and by SLAKE-open. The method matches or beats always-32B-no-think on these sets while running mostly on the cheap model. This heterogeneity—cheap where the small model suffices, expensive only where it must be—is precisely what a uniform single-model strategy cannot express, and is the source of the Pareto gap.

C. Ablations

Gate bake-off. Table V compares escalation signals on the MCQ pool. Cascade quality (accuracy at a fixed compute budget) tracks recoverability-AUROC, not detection-AUROC: the highest-detection gates (learned-rich 0.693, entropy 0.686, max-probability 0.677) are worse cascades than the plain margin, because detecting cheap-model errors is not the same as detecting recoverable ones. The confidence-margin is the best deployable MCQ gate (15.6% escalation to parity); agreement and resolution-stability rank below it and—being family-specific—do not transfer. On open-text, the trained verifier’s confidence is the best gate (detection AUROC 0.85).

Trained verifier and best-of- N . Table VI evaluates the open-text verifier on $n=1,064$ held-out items. Greedy decoding and self-consistency give 0.41; a same-split single 32B pass gives 0.46; the trained verifier’s best-of- N selection gives 0.50, capturing 49% of the oracle gap (seed 0; 35% on seed 1). The verifier’s selection thus matches or beats the $5\times$ -larger 32B (+0.039, 95% CI $[+0.010, +0.066]$ on seed 0; a tie on seed 1)—the honest claim is “competitive with the 32B.”

Adaptive sample count. The optimal-stopping controller (§IV) holds this accuracy at fewer draws (Table VII). Across the three open-text benchmarks it cuts the mean draw count from 8 to 4.3 and the free-text sampling cost from 16.2 to 10.9 FLOP-equivalents (-33%) at held-out iso-accuracy, allocating more draws to the harder PathVQA-open (5.5) than to the easier SLAKE-open (3.5). Because the draws are sequential this trades some batch-1 wall-clock for the FLOP and energy saving; the win is compute, not latency.

Accuracy levers. The confidence-advantage fusion improves PMC-VQA by +0.0135 over always-32B-no-think (95% CI $[+0.010, +0.017]$, $n=33,430$)—a genuine broad-slice win on the largest benchmark, where the two models are comparably skilled with de-correlated errors. The certified veto captures 70% of that gain (+0.0095, CI $[+0.0071, +0.0118]$) while flipping the accuracy-max variant from $1.25\times$ to $0.93\times$ compute (vetoed cells skip the 32B). The team-optimal open-text deferral raises

TABLE III

Main result. Pooled sample-weighted accuracy, compute (FLOP-eq $\times$ one 7B forward), batch-1 latency (concurrent), and energy on the full suite ($n=42,224$, held-out, 10^4 -bootstrap). Δ_{think} vs. always-32B-think; Δ_{oracle} vs. oracle mode-selection. The 32B-think open-text accuracy is now directly measured (not estimated), so Δ_{think} is a fully-measured quantity; the compute-lean, accuracy-max, and fusion points all carry paired-bootstrap 95% CIs. Both method points dominate always-32B-think and oracle-mode-32B on every axis.

System	Acc. (sw)	FLOP -eq	rel. FLOP	Lat. (ms)	Energy (J)	Δ_{think} (95% CI)	Δ_{oracle} (95% CI)
always-7B (cheap floor)	0.5549	1.00	0.22 $\times$	347	45.8	—	—
always-32B-think (naive)	0.5591	4.57	1.00 $\times$	10,522	2,002	—	−0.0139
always-32B-no-think	0.5729	4.57	1.00 $\times$	665	127	—	−0.0001
oracle mode-select 32B	0.5730	4.57	1.00 $\times$	860	164	—	0.0000
Ours: compute-lean	0.5741	2.25	0.49 $\times$	469	84	+0.0150 [+0.011, +0.019]	+0.0010 [−0.003, +0.005]
Ours: accuracy-max	0.5836	4.26	0.93 $\times$	731	137	+0.0245 [+0.022, +0.027]	+0.0106 [+0.009, +0.013]
accuracy-max ⁺ (fusion)	0.5862	5.71	1.25 $\times$	668	177	+0.0271 [+0.024, +0.031]	+0.0131 [+0.010, +0.016]

TABLE IV

Per-benchmark policy and accuracy (compute-lean), against the three single-model baselines, with the paired-bootstrap 95% CI of the method’s delta over the always-32B-no-think reference (all cells real per-sample). “policy”: cheap $\rightarrow$ 32B-nt margin cascade (MCQ) or best-of- N +verifier $\rightarrow$ 32B-nt (open). The cascade matches 32B-no-think within noise on the perception MCQ sets, loses a little only on near-chance MedXpertQA, and wins significantly on PathVQA-open. n as in Table I.

Benchmark	policy	7B	32B-nt	32B-th	ours	Δ vs 32B-nt	95% CI
PMC-VQA	cascade	0.543	0.552	0.549	0.551	−0.0010	[−0.006, +0.004]
SLAKE-cl	cascade	0.825	0.859	0.864	0.852	−0.0072	[−0.024, +0.008]
VQA-RAD-cl	cascade	0.781	0.853	0.841	0.833	−0.0199	[−0.040, +0.000]
PathVQA-cl	cascade	0.841	0.889	0.889	0.888	−0.0009	[−0.004, +0.002]
MedXpert-MM	cascade	0.262	0.307	0.304	0.301	−0.0060	[−0.012, −0.001]
SLAKE-open	bo- N +V	0.736	0.819	0.679	0.809	−0.0093	[−0.036, +0.017]
VQA-RAD-open	bo- N +V	0.465	0.600	0.545	0.595	−0.0050	[−0.060, +0.050]
PathVQA-open	bo- N +V	0.324	0.376	0.109	0.452	+0.0760	[+0.056, +0.096]

TABLE V

Gate bake-off, MCQ pool. Detection AUROC (is the cheap model wrong?) vs. recoverability AUROC (will the 32B fix it?), and escalation to reach 32B-no-think parity. Cascade quality follows recoverability, not detection; the deployed margin gate is best-deployable.

Gate	AUROC _{det}	AUROC _{rec}	esc.%
margin (deployed)	0.725	0.587	15.6
self-verify $P(\text{True})$	0.643	0.614	20.3
max-probability (MSP)	0.677	0.553	20.3
neg-entropy	0.686	0.515	—
learned (grad. boosting)	0.693	0.533	—
agreement (cheap vs. strong)	0.657	—	20.0
resolution-stability	0.724	—	15.5

AUROC_{det}: cheap model wrong? AUROC_{rec}: will 32B fix it? esc.%: escalation to reach 32B-no-think parity.

PathVQA-open to +0.086 over always-32B-no-think (CI [+0.064, +0.106]) and repairs the two slices where the parity gate tied.

D. The fundamental limits

The recoverability bound. Six independent methods (Table VIII) fail to certify any new MCQ slice beyond PMC-VQA—the one slice where the two models are comparably skilled with de-correlated errors—at which the cheap model can safely replace or improve on the 32B. This includes an automatic slice-discovery search that scans 106 candidate slices per split and certifies zero genuinely-

TABLE VI

Open-text best-of- N selection ($n=1,064$ held-out). “gap” = fraction of the greedy $\rightarrow$ oracle gap captured. The trained verifier beats self-consistency and a single same-split 32B pass.

Selector	Accuracy	gap captured
greedy (1 sample)	0.413	0%
self-consistency (majority)	0.411	—
32B single pass (same split)	0.462	—
trained verifier (bo- N)	0.501	35–49%
oracle@ N (upper bound)	0.592	100%

TABLE VII

Adaptive best-of- N (optimal stopping) vs. a fixed $N=8$, per open-text benchmark. Mean draws and free-text sampling cost (FLOP-eq) fall by 33% pooled at held-out iso-accuracy.

Benchmark	$\bar{N}$	$F_{N=8}$	F_{adap}	saved	acc.
SLAKE-open	3.45	16.6	7.6	−54%	0.809
VQA-RAD-open	3.91	16.3	8.4	−48%	0.595
PathVQA-open	5.48	16.0	12.6	−21%	0.452
pooled	4.28	16.2	10.9	−33%	0.563

new beatable slices, below its own label-permutation null (1.6 raw certifications vs. 5.6 under shuffled labels). A k -nearest-neighbor recovery gate also fails to beat the scalar margin on all five datasets. The common cause is error correlation: the two models miss the same questions, so

TABLE VIII

The recoverability bound, confirmed six independent ways. No method certifies a beat-32B MCQ slice beyond PMC-VQA (the one comparable-skill slice); “new cells” counts certified beatable slices beyond PMC.

Method	PMC beat	new cells
confidence-advantage fusion	+0.0135	0
certified confidence veto	+0.0095	0
Bayesian model averaging	+0.0116	0
contrastive decoding	0.000	0
full-posterior logit fusion	n.s.	0
automatic slice discovery (106/split)	—	0

“will the 32B fix this?” carries only ≈ 0.6 AUROC of signal regardless of how it is estimated. This is why “a better gate” does not exist as a lever—the ceiling is a property of the model pair, not of the estimator.

The selection bound. Symmetrically, verification is capacity-bounded but the ceiling is not a size artifact. On a pooled open-text selection task, a 7 $\times$ -larger zero-shot 32B verifier (0.480) ties the small trained 7B verifier (0.475; $\Delta + 0.005$, CI $[-0.023, +0.032]$, not significant), whereas the pure-capacity contrast (32B vs. 7B zero-shot) is significant ($+0.067$). Even the 32B verifier leaves an oracle-to-selection gap of 0.19. Training a small verifier, not enlarging it, is the effective lever—and it is exactly what breaks the selection floor on free text where every training-free selector is stuck.

VI. Discussion and Limitations

We are deliberately explicit about scope. (i) The accuracy edge over the 32B is concentrated on PMC-VQA (the largest slice, via fusion) and on open-text (via the verifier); on the small radiology/pathology MCQ sets the method matches, not beats, the 32B, and the guardrail keeps it there. (ii) The verifier is in-domain for the open-text families it selects on; cross-family transfer is positive but weaker (25% of the oracle gap on a from-scratch MedVLThinker verifier vs. 49% for Lingshu), and the “beats-the-32B” framing is seed-dependent. (iii) Best-of- N is Pareto-dominated on FLOPs when—as here—the strong leg is cheap on both axes; its accuracy win on free text is real, but the deployable efficiency win is the router, and this verdict would change for an expensive/slow strong model. (iv) The optimal-stopping draws are sequential, so on free text the controller wins FLOPs and energy but is slower at batch 1 than a batched fixed- N . (v) All findings are on one model family for the headline; finding 1 generalizes across five, findings 2–3 in direction across two. The method always answers; no result relies on abstention.

Deployment implications. The practical reading is that a hospital deploying a medical VLM need not choose between a fast-but-weak 7B and an accurate-but-slow 32B: a format-aware router captures the accuracy of the large model at a fraction of its cost, with a single knob to trade the residual compute for a small accuracy gain, and a per-

benchmark guardrail that provably never underperforms the strong single model on a covered slice. The components are individually simple—a margin threshold, a batched verifier score, a certified veto—and add no new model to serve beyond the two that a cascade already hosts. The negative results are equally actionable: effort spent engineering a cleverer escalation signal is misdirected, because the recoverability ceiling is a property of the model pair; the returns are in the structure (format-aware actions, an intermediate no-think tier) and in a small amount of verifier training. A natural next step is a slow think tier that pays off on a genuinely reasoning-heavy stream, and a verifier trained across more families to widen the free-text gains.

VII. Conclusion

In medical VQA, the accuracy–cost tension is not a law but a consequence of allocating test-time compute uniformly. Reasoning should fire only where it helps, sampling only where the format rewards it, and escalation only where the large model can actually recover the answer. A format-aware adaptive cascade that follows these rules lets a 7B model Pareto-dominate every fixed way of using a 32B reasoning model—matching its accuracy at half the compute and a twentieth of the latency and energy, and exceeding it, at lower compute, where a trained verifier and a certified fusion apply. The negative half of the story is as useful as the positive: the escalation and selection signals are fundamentally bounded, so the leverage is structural and in a little training, not in a cleverer gate.

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